

MICROCONTROLLER DESIGN

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ The documentation portfolio is present
- ☐ The device (product) is present
- ☐ ENTRY NOT EVALUATED

DOCUMENTATION PORTFOLIO (30 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio is missing one (1) or two (2) components and/or is loosely organized.	Portfolio has all required components in order and is well organized.	
Project Documentation (X1)	The project documentation is inadequate, and/or very few credible sources are referenced.	The project documentation is adequate, and mostly credible sources are included.	The project documentation is comprehensive, and credible resources are included.	
Work Log (X1)	Log is poorly organized and/or incomplete.	Log is adequately detailed and organized and contains most of the required components.	Log is well documented and contains all the required components.	
DOCUMENTATION PORTFOLIO SUBTOTAL (30 points)				

DEVICE/PRODUCT (80 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Complexity (X2)	Product lacks complexity; it includes little or no code, and/or circuit design and control technology complexity.	The product exhibits some degree of complexity; it includes code, circuit design, and control technology complexity.	The product is complex and highly functional; it includes code, circuit design, and control technology complexity.	
Creativity (X2)	The product lacks creativity; very little original thought in developing the project is evident.	Some elements of creativity are expressed; the product is somewhat original.	The work exudes creativity; the product is highly original.	
Technical Skill (X2)	Little technical skill is exhibited in the code and circuit design.	A beyond-basic degree of technical skill is exhibited in the code and circuit design.	A level of mastery of coding and circuit design is exhibited.	

DEVICE/PRODUCT (80 points) – continued				
Connection to the Theme (X1)	Product does not have any, or has very little, connection to the theme.	The product has adequate connection to the theme.	The product has high connection to the theme.	
Design Principles (X1)	Product demonstrates little to no use of design principles.	Product demonstrates adequate use of design principles.	Product demonstrates exceptional use of design principles.	
DEVICE/PRODUCT SUBTOTAL (80 points)				

PRESENTATION (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Organization (X1)	Participant(s) seem unorganized and unprepared for the presentation; the presentation is illogical.	Participant(s) are generally prepared for the presentation; explanation of the product is communicated and generally organized.	The presentation is logical, well organized, and easy to follow; explanation of the product is communicated in an organized and concise manner.	
Articulation (X1)	Communication of the product functionality and design process is unclear, unorganized, and or illogical; leadership and/or 21 st century skills are not evident.	Communication of the product functionality and design process is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the product functionality and design process is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.	
Product Demonstration (X2)	Participant(s) is/are unable to successfully demonstrate the product, and/or the product does not work, or barely works, as intended.	Participant(s) is/are able to partially demonstrate the functionality of the product; the product somewhat works as intended.	Participant(s) is/are successful and effective in the product demonstration; the product works exactly as intended.	
PRESENTATION SUBTOTAL (40 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.	
Indicate the rule violated: _____	
Time violation (a deduction of five [5] points total will be incurred for exceeding the five [5]-minute presentation time limit). Record the deduction.	

PRELIMINARY SUBTOTAL (150 points)	
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To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.	TOTAL (150 points)	
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Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

